T21: Composition for the reported HPM

C1 and discovery routes (a)–(f)

Astra lead; reviewed by Sol 6.1 x2

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proved (Astra lead; reviewed by Sol 6.1 x2); not adopted. This proof establishes the conditional statement in `STATEMENT.md`. The posterior concentration input H03, the addition input H10, and the terminal repair input H24 are hypotheses, not previously proved theorems. No numerical experiment is used.

1 Setting and the composition claim

All notation and hypotheses are as in `STATEMENT.md`. In particular, $\Omega = \{S \subseteq [p] : |S| \le K\}$, $K = \min(p, \lfloor n/2 \rfloor)$, $S_0 \in \Omega$, and

$$F(S) = \log\{\varpi(S)m(S)\}, \qquad Z = \sum_{S\in\Omega} e^{F(S)}, \qquad \pi(S) = e^{F(S)}/Z, \qquad \delta_1 = 1 - \pi(S_0).$$

The score formula and the capped prior are those written out in the statement. The data law is P_0; $\mathbb{P}$ adds the independent algorithm randomness. The total HPM order first compares scores, then (only for exact equality) smaller size and lexicographic order. Its maximum on Ω is $S^\dagger$. The cumulative set of scored models is $\mathcal{S}$, with pre-final subset $\mathcal{S}_{\mathrm{cal}}$. On U^c, where a usable output exists, the reported $\widehat{S}$ is the maximum of $\mathcal{S}$ in that same order.

The probabilities below are never conditioned on passing future diagnostics or on future completion. In route (a), a launched final run executes all its designated draws; in (d), the specified scan is completed; in (e), the specified terminal rounds execute. These are the operational hypotheses in the statement. If external interruption truncates a launched run, the full-count formula of (a) is not asserted for that run.

Theorem 1 (C1)**.** *For every valid problem and every such joint law,*

$$\mathbb{P}\{U \text{ or } \widehat{S} \neq S_0\} \le \mathbb{P}(U) + 2E_0\delta_1 + \mathbb{P}\{U^c, S_0 \notin \mathcal{S}\}. \tag{1}$$

Under H03, $2E_0\delta_1 \to 0$*. Consequently, if* $\mathbb{P}(U) \to 0$ *and the discovery term tends to zero under one of the conditions established below,* $\mathbb{P}\{U^c, \widehat{S} = S_0\} \to 1$*.*

Proof. On $\{\pi(S_0) > 1/2\}$, every $S \neq S_0$ satisfies

$$\pi(S) \le 1 - \pi(S_0) < \pi(S_0).$$

Thus S_0 is the strict score maximum, with no tie. If it has also been scored and the output is usable, the method must return it. Hence

$$\{U^c, \widehat{S} \neq S_0\} \subseteq \{\delta_1 \ge 1/2\} \cup \{U^c, S_0 \notin \mathcal{S}\}.$$

The first event depends only on the data, and $\mathbb{P}(\delta_1 \geq 1/2) = P_0(\delta_1 \geq 1/2) \leq 2E_0\delta_1$. Add $\mathbb{P}(U)$ to get (1).

H03 gives $\delta_1 \to 0$ in P_0-probability, since $\sqrt{\log n/n} \to 0$. For every $\epsilon > 0$, using $0 \leq \delta_1 \leq 1$,

$$E_0\delta_1 \leq \epsilon + P_0(\delta_1 > \epsilon).$$

Taking a limsup, then $\epsilon \downarrow 0$, proves the asserted mean limit. This does not assert an $O(\sqrt{\log n/n})$ frequentist error rate. □

This argument also proves directly the ledger fact needed here: if the global tie-broken maximum has been scored, the reported maximum on a usable run equals it. No unproved clause of T11 or T03 is needed for this deterministic fact.

2 Route (a): fresh final draws

Let $\mathcal{H}_{\text{cal}}$ contain the data and the entire pre-final history, including the tables and calibration streams. The event $\mathcal{F}$ that calibration permits the final SMC run, its tier, grid, T, and N_e are measurable there. On $\mathcal{F}$, the number of initial draws plus independence proposals is

$$Q_f = 4N_e + 4N_e(T + L + 4) = 4N_e(T + L + 5). \tag{2}$$

Off $\mathcal{F}$ set $Q_f = 0$ in indicator-weighted formulas.

The wave-1 result T11(iii) says that, given data and frozen calibration, these designated draws are iid π_0, and their fixed-model miss probability is $(1-\pi_0(S^*))^{Q_f}$. Its proof gives the same assertion conditional on the entire pre-final history: see the collected proof's proposition on the final run as a reading scheme, which fixes all calibration streams before reading fresh final streams. For completeness, the relevant reasoning is as follows. CAL and EX-S leave the final streams independent of $\mathcal{H}_{\text{cal}}$. Each designated call uses unread uniforms and has conditional law π_0 given all preceding reads. Repeated conditioning therefore factors the joint law of its Q_f outputs, even though other random operations occur between calls. Their number is fixed by the pre-final history, not selected by their outcomes. Thus, writing $\mathcal{D}_f$ for the set of these draws,

$$\mathbb{P}(S_0 \notin \mathcal{D}_f \mid \mathcal{H}_{\text{cal}}) = (1 - \pi_0(S_0))^{Q_f} \quad \text{on } \mathcal{F}. \tag{3}$$

Here the true support is fixed by the data-generating problem and in particular does not depend on final randomness. Initial draws and independence proposals are all scored, so $\mathcal{S}_{\text{cal}} \cup \mathcal{D}_f \subseteq \mathcal{S}$ on $\mathcal{F}$. Splitting according to $\mathcal{F}$, then applying (3) and the tower property, gives

$$\begin{aligned}\mathbb{P}(S_0 \notin \mathcal{S}) \leq\ & \mathbb{P}(\mathcal{F}^c, S_0 \notin \mathcal{S}_{\text{cal}}) \\ & + \mathbb{E}\left[\mathbf{1}_{\mathcal{F}}\mathbf{1}_{\{S_0 \notin \mathcal{S}_{\text{cal}}\}}(1 - \pi_0(S_0))^{Q_f}\right].\end{aligned} \tag{4}$$

No independence between calibration success and its earlier scoring events was used. The indicator of earlier absence is measurable before final randomness. Since $1 - x \leq e^{-x}$, the second line is at most the same expectation with $e^{-Q_f\pi_0(S_0)}$.

For any fixed $M > 0$,

$$\mathbb{E}[\mathbf{1}_{\mathcal{F}}e^{-Q_f\pi_0(S_0)}] \leq \mathbb{P}(\mathcal{F}, Q_f\pi_0(S_0) \leq M) + e^{-M}.$$

The stated divergence on launched runs therefore makes this expectation vanish, by first taking $n \to \infty$ and then $M \to \infty$. Together with the nonlaunch-and-no-earlier-hit term, this proves (a). This use of a fixed exponent does not apply to the adaptive calibration draw count.

One-deletion start mass and actual budgets

Assume $s_0 \geq 1$, $i \in S_0$, and $\widetilde{S}_0 = S_0 \setminus \{i\}$ has π_0-mass at least $c > 0$. Its being the start mode is allowed but not needed for the following inequality. The addition is feasible because $s_0 \leq K$. Hypothesis H10 gives

$$\pi_0(S_0) \geq c r_{s_0-1} = \frac{c s_0}{(2p - s_0)\sqrt{n+1}}.$$

Consequently, on $\mathcal{F}$,

$$Q_f \pi_0(S_0) \geq \frac{4 c s_0 N_e (T + L + 5)}{(2p - s_0)\sqrt{n+1}}. \tag{5}$$

Since $p \geq s_0 \geq 1$, $p \leq 2p - s_0 \leq 2p$. For fixed s_0, c, the expression on the right has order $N_e(T + L + 5)/(p\sqrt{n})$, with constants independent of the tier and grid. This is the order of a guaranteed lower bound, not a two-sided order for the actual intensity. If $p \leq C\sqrt{n}$, then $N_e \geq n$, $T \geq 1$, and

$$\frac{4 c s_0 N_e (T + L + 5)}{(2p - s_0)\sqrt{n+1}} \geq \frac{2 c s_0}{C} \sqrt{\frac{n}{n+1}} (L + 6) \longrightarrow \infty.$$

This includes p proportional to $\sqrt{n}$, since $L \geq \lceil \log_2(n+1) \rceil$. If the mass hypothesis holds only on an event G_n with $\mathbb{P}(G_n) \to 1$, split the expectation in (4) over G_n and G_n^c; the extra term is at most $\mathbb{P}(G_n^c)$. A lower bound which ceases to diverge gives no upper bound on the true intensity and no proof of failure outside this sufficient range. No conclusion of T27 is invoked.

3 Route (b): a Gibbs execution

At the specified execution put $A = X_{\mathrm{pre}-j} \setminus \{j\} = S_0 \setminus \{j\}$. The two candidate configurations are A and $A \cup \{j\}$. If $j \in S_0$, the latter is S_0 and is feasible because $s_0 \leq K$. If $j \notin S_0$, the former is S_0 and is feasible, even if the alternative exceeds the cap. In either case SR scores S_0. This is true whether or not the ensuing Gibbs draw selects it. Thus a missed truth implies the absence of the specified execution, and $\mathbb{P}(U^c, S_0 \notin \mathcal{S}) \leq \mathbb{P}(U^c, E_G^c)$.

The state immediately before coordinate j may have been changed by any earlier labels in the random-order round. Therefore a condition solely on round-start occupancy cannot be substituted for the displayed equality at execution time. No round-end occupancy claim is being made.

4 Route (c): stopped swap opportunities

Suppose the current proxy has the form $S = (S_0 \setminus \{l\}) \cup \{j\}$, where $l \in S_0$, $j \notin S_0$, and $k = |S| = s_0 \in \{1, \ldots, p-1\}$. There is exactly one deletion/insertion pair producing S_0, namely (j, l). Conditional on the full past just before a uniform swap, that pair has probability

$$b = \frac{1}{k} \frac{1}{p - k}. \tag{6}$$

Acceptance is irrelevant because the proposal is scored.

We prove the stopped-opportunity bound directly, rather than citing an unproved ledger clause. Order all executed elementary operations chronologically; the finite budgets permit such an ordering, and stopped histories include the identity of the particle and the next operation. Count only eligible swap executions before the first truth score. Let τ_i be the time of the i-th such execution, or ∞ if it does not occur. Eligibility and the absence of prior discovery are known immediately before each draw. Let A_i be the event that $\tau_i < \infty$ and its proposal misses S_0, and put A_0 equal

to the whole probability space. The event $\{\tau_i < \infty\}$ implies A_{i-1} for $i \geq 2$: a hit at an earlier counted execution would have stopped counting. Summing the conditional miss probabilities over the possible pre-execution histories gives

$$\mathbb{P}(A_i) = (1-b)\mathbb{P}(\tau_i < \infty) \leq (1-b)\mathbb{P}(A_{i-1}) \leq (1-b)^i.$$

This argument does not condition on a future opportunity count. If the whole run misses truth and $O \geq D$, then A_D occurs (for $D = 0$, simply use the bound 1). Hence

$$\begin{aligned}\mathbb{P}(S_0 \notin \mathcal{S}) &\leq \mathbb{P}(S_0 \notin \mathcal{S}, O < D) + \mathbb{P}(S_0 \notin \mathcal{S}, O \geq D) \\ &\leq \delta + (1-b)^D.\end{aligned}$$

The proof remains valid after conditioning on data or initial information for which the stipulated proposal laws hold and D, b are fixed. For PX choose $D_n = \lceil \omega_n k(p-k) \rceil$. Then $(1-b)^{D_n} \leq e^{-bD_n} \leq e^{-\omega_n} \to 0$; the deficient-exposure term vanishes by PX. A relation between two models, such as ONE, alone does not supply this execution count.

For the neighbour-swap proposal, the unique useful pair is still (j, l). Its conditional probability is zero if $l \notin N(j) \setminus S$, and otherwise is

$$b_e = \frac{1}{k\,|N(j) \setminus S|}.$$

The actual graph is essential. For a fixed deterministic $b_* \in (0, 1]$, if $b_e \geq b_*$ at every counted execution, the same conditioning argument yields $\mathbb{P}(A_i) \leq (1-b_*)\mathbb{P}(A_{i-1})$, proving the asserted extension. For a random lower bound measurable at a conditioning boundary, apply this argument conditionally and then average; the resulting unconditional term is the expectation of $(1-b_*)^D$. There is no division by zero: an execution with the required edge has a nonempty insertion set. The uniform-swap formula was restricted to $1 \leq k < p$ precisely to exclude the method's self-loop boundaries.

5 Route (d): tables and completed scans

For $n \geq 4$, $\lfloor \sqrt{n} \rfloor \leq \lfloor n/2 \rfloor$; for $n = 3$, $\lfloor \sqrt{n} \rfloor = 1 = \lfloor n/2 \rfloor$. Also $B \leq p$, so $B \leq K$. Thus every subset of a block is feasible and every completed block table includes its true full-response score. In particular, if $S_0 \subseteq C_b$ and that table is constructed, S_0 has been scored.

The feasible flip-or-swap neighbourhood is symmetric: the reverse of an addition is a deletion, the reverse of a deletion is an addition to the original feasible state, and a swap is reversed by the opposite swap. If a completed scan has centre c in the neighbourhood of S_0, then S_0 lies in the neighbourhood scanned at c. The method scores all of that neighbourhood, including at its last extra scan. This is also the scoring portion of wave-1 T20(ii); it needs no strict-mode assumption, although T20(ii)'s subsequent move/certificate conclusions do. It follows that

$$\{U^c, S_0 \notin \mathcal{S}\} \subseteq \{U^c, E^c_{TS}\}.$$

Once a score has been obtained it remains in the HPM ledger even if the schedule cannot be constructed later. Consequently schedule success is not needed for this route, provided the output is usable. C1 already accounts for output unavailability.

6 Route (e): terminal repair under H24

We prove the conditional repair inequality from the uniform one-round bounds assumed in H24. Fix the data and the entire available past at the selection boundary. On the availability event an eligible particle starts at $x \in \mathcal{R}_d$. Its future terminal rounds use fresh randomness and the same kernel K_1, without intervening resampling. The round count H, and the constants $a = a_n$, $\eta = \eta_n$, are now fixed. H24 states

$$\inf_{x \in \mathcal{R}_d} K_1(x, \{S_0\}) \geq a, \qquad \sup_{x \in \mathcal{R}_d} \mathbb{P}_x(\text{an exit during one round}) \leq \eta. \tag{7}$$

These are assumptions on the actual hybrid round, including all its labels; they are not derived here from leakage or gain conditions.

Let J be the event of any elementary-state exit from $\mathcal{R}_d$ during the H rounds. Before the first exit each round starts in $\mathcal{R}_d$. At each such boundary apply the second inequality in (7) conditional on that boundary's past. Summing the probabilities of a first exit in the different rounds gives

$$\mathbb{P}(J \mid \text{selection past}) \leq H\eta. \tag{8}$$

For $h = 0, \ldots, H$, let B_h mean no exit through round h and none of its first h round-end states equals S_0. With B_0 the whole conditional space, on B_{h-1} the next start is in $\mathcal{R}_d$, so the first inequality in (7) gives

$$\mathbb{P}(B_h \mid \text{selection past}) \leq (1-a)\mathbb{P}(B_{h-1} \mid \text{selection past}) \leq (1-a)^h.$$

Every terminal endpoint is a scored state. Indeed initial particle states are scored; Gibbs updates score both feasible alternatives under SR, swap and independence proposals are scored, and an acceptance moves to a scored state while a rejection retains the current scored state. Resampling before the terminal segment merely copies scored states. Thus a truth endpoint implies a truth score. Let M_H be the event that $S_0 \notin \mathcal{S}$ when those H rounds end. It follows that M_H is contained in $J \cup B_H$, and

$$\mathbb{P}(M_H \mid \text{selection past}) \leq (1-a_n)^H + H\eta_n. \tag{9}$$

If the initial state is already truth, its score is already present and M_H is empty. In particular this case cannot worsen any whole-run bound. More generally, a truth visit before an endpoint only improves discovery. At $H = 0$ use the convention $(1-a)^0 = 1$.

Let A be the past-measurable availability event. Split whole-run failure over A^c and A, use (9), cap the conditional bound by 1, and integrate. This gives

$$\mathbb{P}(S_0 \notin \mathcal{S}) \leq \mathbb{P}(A^c) + \mathbb{E}\left[\mathbf{1}_A \min\{1, (1-a_n)^H + H\eta_n\}\right].$$

For a final-run segment use $A_F = \mathcal{F} \cap \{\text{eligible particle available}\}$. Splitting first into nonlaunch, launch without an eligible particle, and A_F, improves this to

$$\begin{aligned}\mathbb{P}(S_0 \notin \mathcal{S}) \leq{}& \mathbb{P}(\mathcal{F}^c, S_0 \notin \mathcal{S}_{\text{cal}}) + \mathbb{P}(\mathcal{F}, \text{no eligible particle}) \\ &+ \mathbb{E}\left[\mathbf{1}_{A_F} \min\{1, (1-a_n)^H + H\eta_n\}\right].\end{aligned} \tag{10}$$

Selection may depend on all states and randomness observed so far, since (7) is uniform over eligible states. It may not inspect future randomness. At a pilot terminal boundary the same argument uses the pilot's forward filtration: unread uniforms are fresh there. Conditioning instead on the grid eventually produced by that pilot would reveal future information and is not part of this argument.

For arbitrary H, $(1-a_n)^H \le e^{-Ha_n}$; thus the conditions $Ha_n \to \infty$ and $H\eta_n \to 0$ in probability on availability make the capped expectation tend to zero, by the same bounded-variable argument used for C1. For $H = L + 4$, H24's explicit asymptotic inputs give this directly: when s_0, d are fixed and $B \le 16$,

$$c_n = \frac{1}{(s_0 + d + B + 2)!} \ge \frac{1}{(s_0 + d + 18)!} > 0.$$

On an event of probability tending to one, $a_n \ge c_n/2$, while $L \to \infty$; hence $Ha_n \to \infty$ in probability, and $H\eta_n \to 0$ is the other assumed input. The availability and nonlaunch terms must still vanish. The proposed formulas for a_n, η_n in the statement only specify which H24 input is intended; they do not prove H24 or its rates.

7 Route (f): completed exact enumeration

When

$$\sum_{k=0}^{K} \binom{p}{k} \le (5H_0 + 4N_0)p, \tag{11}$$

the method chooses enumeration before constructing the SMC components. On its completion, every feasible model has been scored, so $\mathcal{S} = \Omega$. The running-best rule uses precisely the global tie order and therefore returns $\widehat{S} = S^\dagger$. Its accumulated evidence and inclusion numerators are, under EX-A,

$$Z = \sum_{S \in \Omega} \varpi(S) m(S), \qquad Z_j = \sum_{\substack{S \in \Omega \\ j \in S}} \varpi(S) m(S).$$

Consequently the evidence is exact and the primary PIP is Z_j/Z. No Monte Carlo PIP or secondary-pooling assertion is used. This is finite enumeration, not an application of an unproved asymptotic Monte Carlo theorem. Wave-1 T09(i) establishes the exact start normaliser and sampler when those components are used; it is compatible with the method but unnecessary for these direct enumeration identities.

For fixed $p \ge 1$, the left side of (11) is at most 2^p, whereas $(5H_0 + 4N_0)p \ge 4np \to \infty$. Thus the branch inequality holds eventually (and eventually $K = p$). This proves eventual branch selection, not completion under a fixed external allocation. Conditional on completion, the discovery term in C1 is zero; H03 is still needed for truth consistency.

8 Scope and source locators

The method used is *METHODOLOGY_v3.md*, version 3.0 of 19 October 2026: §3 for budgets; §5 for the posterior; §6.2 for exact enumeration; §6.3 for tables and the start; §6.5 for move proposals; §§6.8–6.9 for calibration and fresh final streams; and §6.10 for the all-scored HPM and local scans. The only cited proved wave-1 statements are in *COLLECTED_PROOFS_v2.tex*: the T11(iii) chapter, its "final run is a reading scheme" proposition and theorem; the T20 chapter, clause (ii-A), first scoring implication; and the T09 chapter, part (i), quoted only for the exact start component. All facts required beyond these and the explicit H03/H10/H24 hypotheses were proved above from the method.

For statistical interpretation, G/R/SE/BM-C are the proposed primitive conditions behind T03, not a proved route to H03 in this document. No EXP or EXP' expansion is invoked and no no-switch conclusion is made; therefore no cluster-separation implication is being presumed. The result keeps operational failure separate from posterior error and discovery. It proves neither evidence nor PIP accuracy on the SMC branch.